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Week 03: Wednesday, AST 5011: Astrophysical Systems

Stellar Energy Sources

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With contributions totally ripped off from Carl Fields (UA), Mike Zingale (SUNY), Cole Miller (UMD), and Abi Nolan (Purdue).

Stellar Energy Sources: Gravitation

Why Gravity Matters (Even Though It Can't Power the Sun Forever)

In previous classes, we discussed possible energy sources that could power stars. What ultimately rules out everything except nuclear fusion is the combination of:

1. The energy per unit mass of the process, and
2. The long-term stability of the process.

Let's quantify these for the Sun.

Energy Budget of the Sun

The Sun's luminosity is

$$L_\odot \approx 3.8 \times 10^{33}\ \mathrm{erg\ s^{-1}}$$

Over its lifetime so far (~4.6 Gyr):

$$E_{\mathrm{tot}} \approx L_\odot \cdot t_\odot \sim 5.5 \times 10^{50}\ \mathrm{erg}$$

With a solar mass

$$M_\odot \approx 2 \times 10^{33}\ \mathrm{g}$$

this corresponds to an energy release per gram of

$$\frac{E}{M} \sim 2.8 \times 10^{17}\ \mathrm{erg\ g^{-1}}$$

Comparison of Energy Sources

Process	Energy per mass (erg g^{-1})	Viable for Sun?
Chemical reactions	$\sim 10^{12}$	Too small
Gravitational contraction	$\sim 10^{15}$	Too small
Nuclear fission	$\sim 10^{18}$	Short-lived / unstable
Nuclear fusion (H → He)	$\sim 10^{20}$	Perfect

> In-class question: Why does both the energy per mass and the stability of the process matter?

▶ Solution

Gravitational Binding Energy

To understand gravity as a source of energy, we compute the gravitational binding energy — the energy required to remove a star's material layer by layer and disperse it to infinity.

For a spherically symmetric star:

$$E_{\text{grav}} = \int_0^R \frac{GM(<r)}{r}\,\rho(r)\,4\pi r^2 dr$$

where $M(<r)$ is the mass enclosed within radius r.

Example: Constant-Density Sphere

If the density is uniform, $\rho(r) = \rho_0$, then

$$M(<r) = \frac{4}{3}\pi\rho_0 r^3$$

and the gravitational energy integral yields

$$E_{\text{grav}} = \frac{3}{5}\frac{GM^2}{R}$$

Note: Real stars are centrally concentrated, so in practice:

$E_{\text{grav}} \sim \frac{GM^2}{R}$ with a larger prefactor for central concentration.

> In-class question: Why does central concentration increase the prefactor?

▶ Solution

Virial Theorem and Stellar Energy

The Virial theorem relates total energy $E_{\rm tot}$, thermal energy U, and gravitational energy $E_{\rm grav}$ in a star:

$2U + E_{\rm grav} = 0 \quad \Rightarrow \quad U = -\frac{1}{2}E_{\rm grav}$

- This explains why only half of the energy released by contraction actually heats the star; the other half is radiated away.
- Combining the Virial theorem with the Kelvin–Helmholtz timescale shows that stars powered purely by contraction would be much more luminous initially and fade rapidly.

In-class Exercise: Gravitational Energy and Central Concentration

We know the gravitational binding energy of a star depends on how mass is distributed:

$$E_G = -\frac{3}{5}\frac{GM^2}{R} \quad \text{(uniform density)}$$

For a centrally concentrated star, the prefactor can be larger in magnitude. We will explore this numerically.

Tasks

1. Consider two idealized density profiles for a star of mass $M = 1M_\odot$ and radius $R = 1R_\odot$:
   - Uniform: $\rho(r) = \frac{3M}{4\pi R^3}$
   - Centrally concentrated: $\rho(r) = \rho_c(1-(r/R)^2)$, with ρ_c chosen so total mass equals M
2. Write a Python function to compute the enclosed mass $M_r(r)$ and the differential gravitational energy:

$$dE_G = -\frac{GM_r(r)\,dm}{r}, \quad dm = 4\pi r^2\rho(r)dr$$

3. Integrate dE_G over $r \in [0, R]$ to compute the total gravitational energy E_G for both density profiles.

4. Compare the prefactors and discuss: which profile is more tightly bound and why?

▶ Solution

The Kelvin–Helmholtz Timescale

For the Sun, the gravitational binding energy is approximately

$E_{\rm grav} \approx \frac{GM_\odot^2}{R_\odot} \approx 3.8\times10^{48}\ {\rm erg}$

Dividing by the solar luminosity gives the Kelvin–Helmholtz timescale:

$$\tau_{\mathrm{KH}} = \frac{E_{\mathrm{grav}}}{L_{\odot}} \approx 3 \times 10^{7}\ \mathrm{yr}$$

This was a major crisis in 19th-century physics because geology indicated that the Earth (and the Sun) must be much older.

In-class question: Why is the KH time the maximum lifetime for a gravity-powered Sun?

▶ Solution

Gravity as a Real Energy Source

Although gravity cannot power the Sun for billions of years, it does dominate many systems:

Protostars

- Energy source: contraction
- Lifetime roughly the Kelvin–Helmholtz time
- Early contraction is rapid; later contraction slows

Free-fall time limit:

$$\tau_{\mathrm{ff}} \sim \sqrt{\frac{R^3}{GM}}$$

Gas Giants (for example, Jupiter)

Internal luminosity:

$$L_J \approx 4.6 \times 10^{24}\ \mathrm{erg\,s^{-1}}$$

Gravitational energy:

$$E_{\mathrm{grav}} \approx 3.7 \times 10^{43}\ \mathrm{erg}$$

Timescale:

$$\tau \approx 2.6 \times 10^{11}\ \mathrm{yr}$$

In-class question: Why can Jupiter still shine gravitationally today?

▶ Solution

Stars, Brown Dwarfs, and Planets

- Stars: hydrogen fusion
- Brown dwarfs: temporary deuterium fusion
- Gas giants: gravitational contraction only

This distinction is fundamentally about interior temperature.

Gravitational Collapse and Supernovae

For massive stars:

- Lifetime fusion energy: roughly a few $\times 10^{51}$ erg
- Neutron star formation:

$$E \sim \frac{GM_{\mathrm{NS}}^2}{R_{\mathrm{NS}}} \sim \text{few} \times 10^{53} \text{ erg}$$

Gravity releases about 100 times more energy than fusion. This energy powers the supernova explosion.

Other Energy Reservoirs $\sim GM^2/R$

- Rotation (for example, pulsars)
- Thermal energy (via the virial theorem)
- Magnetic energy (upper bound)

In-class question: Why does gravity set the maximum scale for so many energy sources?

▶ Solution

Other Non-Fusion Energy Sources

Crystallization

- Energy per ion: roughly kT_m
- Relevant for:
  - Old white dwarfs
  - Terrestrial planets

Low energy, but astrophysically important in specific cases.

Fission

- Energy per mass: roughly $10^{-3}c^2$
- Not viable for stars because:
  - Insufficient heavy elements
  - Critical mass leads to explosive instability

In-class question: Why is stability just as important as energy yield?

▶ Solution

Big Picture

Gravity:

- Sets fundamental energy scales
- Powers protostars, planets, accretion, and collapse
- Explains why fusion is necessary for long-lived stars

Whenever you see GM^2/R in astrophysics, pay attention — it usually matters.

In-class Exercise: Kelvin–Helmholtz Timescale

We want to explore how the Kelvin–Helmholtz (KH) timescale depends on stellar mass and radius, and how it compares to the Sun.

The KH timescale is defined as:

$$\tau_{\rm KH} = \frac{E_{\rm grav}}{L} \sim \frac{GM^2}{RL}$$

where $E_{\rm grav}$ is the gravitational binding energy, M is the stellar mass, R is the stellar radius, and L is the luminosity.

Tasks:

1. Compute $\tau_{\rm KH}$ for a star with the following properties:

Star	M [$M_\odot$]	R [$R_\odot$]	L [$L_\odot$]
Sun	1.0	1.0	1.0
Star A	0.5	0.7	0.2
Star B	2.0	1.5	10.0

2. Plot $\tau_{\rm KH}$ as a function of mass for a range of solar-type stars (0.1–10 $M_\odot$), assuming $R \propto M^{0.8}$ and $L \propto M^{3.5}$. Use a log-log scale.

3. Compare the KH timescale to the main-sequence lifetime. Which stars would shine primarily via gravitational contraction if fusion did not occur?

▶ Solution

Basics of Thermonuclear Fusion

This section introduces the fundamental physical principles behind nuclear fusion in stars.

Nuclear Potential and the Fusion Barrier

Consider two nuclei interacting as a function of their separation r. Define the zero of energy when the nuclei are at rest at infinite separation.

- At large separations, electrostatic repulsion dominates:

$E(r) \sim \frac{Z_1 Z_2 e^2}{r}$

where Z_1 and Z_2 are the proton numbers.

- At very small separations ($r \lesssim 1$ fm), the strong nuclear force becomes dominant and binds the nucleons together. The total potential energy becomes negative.

Fusion occurs if nuclei approach closely enough for the strong force to act.

In-class question

Why does electrostatic repulsion dominate at large distances, while the strong force dominates only at short distances?

▶ Solution

Classical vs Quantum Picture

Classically, fusion requires nuclei to have enough kinetic energy to overcome the Coulomb barrier.

- Typical Coulomb barrier for proton–proton fusion:

$E_{\text{barrier}} \sim 1 \text{ MeV}$

- Equivalent temperature:

$T \sim 10^{10} \text{ K}$

- Core temperature of the Sun:

$T_c \sim 1.5 \times 10^7 \text{ K}$

Clearly, classical physics alone cannot explain fusion in stars.

Quantum Tunneling

Quantum mechanics allows particles to tunnel through potential barriers. The transmission probability for a particle of energy E through a barrier $V(x)$ is:

$$T(E) = \exp\left[-2 \int dx \sqrt{\frac{2m}{\hbar^2}(V(x) - E)}\right]$$

In-class question

Why does tunneling allow fusion at temperatures far below the classical barrier temperature?

▶ Solution

Density-Driven Fusion and the Fermi Energy

From the uncertainty principle:

$$\Delta x \, \Delta p \gtrsim \hbar$$

For number density n:

$$\Delta x \sim n^{-1/3}$$

Fermi momentum:

$$p_F \sim \hbar n^{1/3}$$

Fermi energy (non-relativistic):

$$E_F = \frac{p_F^2}{2m}$$

Cross Sections and Reaction Rates

The reaction rate per unit volume:

$$R = \sigma(v) \, v \, n_1 n_2$$

For a thermal distribution:

$$\langle \sigma v \rangle = \left(\frac{8}{\pi m}\right)^{1/2} (kT)^{-3/2} \int_0^\infty \sigma(E) E e^{-E/kT} \, dE$$

The Gamow Peak

For non-resonant reactions:

$\sigma(E) = \frac{S(E)}{E} e^{-2\pi\eta}$

with

$\eta = \frac{Z_1 Z_2 e^2}{\hbar v}$

The competition between tunneling and thermal suppression produces the Gamow peak.

In-class question

Why does a small change in temperature significantly shift the Gamow peak and fusion rate?

▶ Solution

Temperature Sensitivity of Fusion Rates

Because fusion rates depend exponentially on temperature, they are often approximated as power laws:

$$\epsilon = \epsilon_0 \, \rho^{\lambda} T^{\nu}$$

where:

- λ describes density sensitivity
- ν is often very large (10–40)

This explains:

- Strong thermal regulation in stars
- Long main-sequence lifetimes
- Short, violent late burning stages

In-class question

Why does a large value of ν help stabilize stars against collapse or runaway heating?

▶ Solution

Nuclear Binding Energy and Stellar Power

The binding energy per nucleon:

- Increases from hydrogen to iron
- Peaks near ^{56} Fe
- Decreases for heavier nuclei

Consequences:

- Fusion of light nuclei releases energy
- Fusion beyond iron is endothermic

Most stellar energy comes from:

$$^{1}\mathrm{H} \rightarrow ^{4}\mathrm{He}$$

This releases $\sim 7\mathrm{MeV}$ per nucleon, far more than later fusion stages.

In-class question

Why do stars spend most of their lives burning hydrogen rather than heavier elements?

▶ Solution

In-class Exercise: Temperature Dependence of Fusion Rates

In stars, the nuclear fusion rate per particle pair can be approximated as:

$$R(T) \propto T^{\nu} \exp\left(-\frac{b}{T^{1/3}}\right)$$

where:

- T is the temperature in K,
- ν is a power-law exponent (from cross-section behavior),
- b is related to the Gamow energy and Coulomb barrier.

Tasks

1. Write a Python function to compute the normalized fusion rate $R(T)$ for given T, ν, and b.
2. Plot $R(T)$ versus T on a logarithmic temperature scale (10^6 K to 10^8 K) for:
   - $\nu = 4$ and $b = 20$
   - $\nu = 6$ and $b = 20$
3. Compare how sensitive $R(T)$ is to small changes in T for the two different ν values.
4. Identify the approximate "Gamow peak" region where $R(T)$ is largest.

▶ Solution

Specific Fusion Reactions

Overview

In the following, we move from general principles of fusion to specific fusion reactions relevant for stars.
As discussed previously, the overwhelming majority of nuclear binding energy release occurs when hydrogen is fused into helium-4, so we begin there.

Hydrogen to Helium: Global Reaction

A naive way to write hydrogen fusion is:

$$4^1\mathrm{H} \rightarrow^4 \mathrm{He}$$

Charge conservation requires electrons, and lepton number conservation requires neutrinos:

$$4^1\mathrm{H} + 2e^- \rightarrow^4 \mathrm{He} + 2\nu_e$$

Total energy released:

$$Q = 26.73 \text{ MeV}$$

Neutrinos escape the star, so their energy does not contribute to pressure support.

In-class question

Why must this reaction occur through multiple stages rather than a single step?

▶ Solution

Why Hydrogen Fusion Is Slow

1. Six particles cannot realistically interact simultaneously.
2. Two protons must convert into neutrons → weak interaction, which is slow.

This weak interaction bottleneck is why stars live for billions of years.

Proton–Proton (p–p) Reaction

The first step of hydrogen burning is:

$$^1\mathrm{H} +^1 \mathrm{H} \rightarrow^2 \mathrm{H} + e^+ + \nu_e$$

Energy released (including positron annihilation):

$$Q \approx 1.442 \text{ MeV}$$

Reaction rate (per volume)

$$r_{p\text{-}p} = \frac{1.15\times10^9}{T_9^{2/3}} X^2 \left(\frac{\rho}{1\ \mathrm{g\,cm^{-3}}}\right)^2 e^{-3.38/T_9^{1/3}} \quad \mathrm{cm^{-3}\,s^{-1}}$$

where $T_9 = T/10^9\ \mathrm{K}$.

In-class question

Why does the rate scale as $X^2\rho^2$?

▶ Solution

Energy Generation Estimate (Solar Core)

Typical solar core values:

- $T = 1.5\times10^7\ \mathrm{K}$
- $\rho = 100\ \mathrm{g\,cm^{-3}}$
- $X = 0.7$

This gives:

- $\sim 10^6$ reactions g^{-1} s^{-1}
- $\epsilon \sim 2\ \mathrm{erg\,g^{-1}\,s^{-1}}$

In-class question

Compare this with chemical burning. Why are stars still powerful?

▶ Solution

Temperature Sensitivity

The effective temperature exponent:

$$\nu_{p\text{-}p} = \frac{11.3}{T_6^{1/3}} - \frac{2}{3}$$

At solar conditions: $\nu \approx 4$

Lesson: weak-interaction–limited reactions are weakly temperature sensitive.

Formation of Helium-3

Once deuterium forms, it rapidly reacts:

$$^1\mathrm{H} +^2\mathrm{H} \rightarrow^3\mathrm{He} + 5.493\ \mathrm{MeV}$$

This reaction is fast (no weak interaction).

Proton–Proton Chains

Helium-3 can form helium-4 via three branches:

- pp-I
- pp-II
- pp-III

Effective energy generation rate:

$$\epsilon_{\rm pp} \approx \frac{2.4\times10^4\,\rho X^2}{T_9^{2/3}} e^{-3.38/T_9^{1/3}} \quad \mathrm{erg\,g^{-1}\,s^{-1}}$$

CNO Cycle

At higher temperatures, hydrogen burns via the CNO cycles.

Example (CNO-I):

$$^{12}\mathrm{C} \to \cdots \to ^{12}\mathrm{C} + ^4\mathrm{He}$$

This is a catalytic cycle.

Energy generation rate

$$\epsilon_{\rm CNO} \approx \frac{4.4\times10^{25}\,\rho XZ}{T_9^{2/3}} e^{-15.2/T_9^{1/3}} \quad \mathrm{erg\,g^{-1}\,s^{-1}}$$

Temperature sensitivity

$$\nu_{\rm CNO} = \frac{50.8}{T_6^{1/3}} - \frac{2}{3}$$

At $T \sim 2 \times 10^7$ K: $\nu \sim 18$

In-class question

Why does this lead to convective cores in massive stars?

▶ Solution

Helium Burning: Triple-Alpha Process

Direct ^{4}He + ^{4}He fusion fails due to unstable ^{8}Be.

Instead:

$$3^4\mathrm{He} \rightarrow^{12}\mathrm{C}$$

Requires:

- $T \gtrsim 10^8\ \mathrm{K}$
- No weak interactions → very fast

Energy generation

$$\epsilon_{3\alpha} = \frac{5.1 \times 10^8\, \rho^2 Y^3}{T_9^3} e^{-4.4/T_9} \quad \mathrm{erg\,g^{-1}\,s^{-1}}$$

Temperature exponent

$$\nu_{3\alpha} = \frac{4.4}{T_9} - 3$$

At 10^8 K: $\nu \sim 40$

Advanced Burning Stages

- Carbon burning
- Oxygen burning
- Silicon burning
- Photodisintegration
- Nuclear statistical equilibrium

At these temperatures:

- Neutrino losses dominate
- Energy generation becomes inefficient

Summary

- Hydrogen burning: pp chains (low T), CNO cycles (high T)
- Helium burning: triple-alpha
- Advanced burning: increasingly complex, short-lived
- Neutrinos: dominate energy loss at high T

The nuclear binding energy curve explains stellar lifetimes.

In-class Exercise: Hydrogen Fusion Rates

In this exercise, we explore the proton-proton (p–p) reaction rate in the Sun and compare it with the CNO cycle. You will use Python to compute and visualize how reaction rates depend on temperature and composition.

Background

The p–p reaction rate per volume is approximately:

$$r_{p\text{-}p} = \frac{1.15 \times 10^9}{T_9^{2/3}} X^2 \left(\frac{\rho}{1\,\mathrm{g\,cm^{-3}}} \right)^2 e^{-3.38/T_9^{1/3}} \quad \mathrm{cm^{-3}\,s^{-1}}$$

where:

- $T_9 = T/10^9\,\mathrm{K}$
- X = hydrogen mass fraction
- ρ = density (g cm^{-3})

The CNO cycle rate is:

$$r_{\mathrm{CNO}} \propto \rho X Z T_9^{-2/3} e^{-15.2/T_9^{1/3}}$$

where Z is the metallicity.

Tasks

1. Plot the p–p reaction rate as a function of temperature T for a fixed solar core density $\rho = 100\,\mathrm{g\,cm^{-3}}$ and hydrogen fraction $X = 0.7$.
2. Overplot the CNO reaction rate for the same density and X, assuming $Z = 0.02$.
3. Identify the temperature at which the CNO cycle overtakes the p–p chain.
4. Plot the effective temperature exponent $\nu = d \ln \epsilon / d \ln T$ for both reactions.

Questions

1. Which reaction dominates at solar core temperatures? At what temperature does the dominance switch?
2. Why is the p–p reaction weakly temperature sensitive while the CNO cycle is highly temperature sensitive?
3. How does the temperature sensitivity affect convective stability in massive stars?
4. Why must hydrogen fusion proceed through multiple stages?
5. Why do stars release vastly more energy per reaction than chemical reactions?

▶ Solution

Thermonuclear Energy Sources

Charged Particle Thermonuclear Reactions

Consider a thermonuclear reaction of the form

$$\alpha\ (\text{"projectile"}) + X\ (\text{"target nucleus"}) \rightarrow Z^*\ (\text{compound nucleus}) \rightarrow Y + \beta\ ($$

or often written as

$$X(\alpha, \beta)Y$$

where the left handside is called the **"entrance channel"**, Z^* is an intermediate (almost always) excited state as a result of the reaction, that leads to the **"exit channel"** and the corresponding **products**.

An example reaction

Consider the following thermonuclear reaction:

$$p +^{11}\mathrm{B} \rightarrow^{12}\mathrm{C}^*$$

this compound nucleus can then break up into a various number of products, or exit channels that include,

$$\begin{aligned}
{}^{12}\mathrm{C}^* &\rightarrow^{12}\mathrm{C} + \gamma && (1)\\
&\rightarrow^{11}\mathrm{B} + p && (2)\\
&\rightarrow^{11}\mathrm{C} + n && (3)\\
&\rightarrow^{12}\mathrm{N} + e^- + \bar{\nu}_\mathrm{e} && (4)\\
&\rightarrow^{8}\mathrm{Be} + \alpha && (5)
\end{aligned}$$

where (**) is another excited (or ground) state of ^{12}C, γ is photon, p is proton, n is neutron, e^- is electron, $\bar{\nu}_\mathrm{e}$ is an anti-electron type neutrino, and α is a ^{4}He nucleus.

Nuclear Energetics

We can begin by defining the **total binding energy** as the energy required to break up and disperse to infinity all the constituent nucleons (protons and neutrons) in that nucleus.

$$B_\epsilon = (\text{mass of constituent nucleons} - \text{mass of bound nucleus})\ c^2 (\text{MeV})$$

similarly, we can define the **average binding energy per nucleon**, B_ϵ/A, where A is the total nucleon mass number, A (in integer amu).

> This is often used as a measure of the energy required to remove the most energetic nucleon from a given nucleus in its ground state.

Average Binding Energy Per Nucleon

binding energy

Credit: Data from Wapstra et al. (1988) - Plotted is the binding energy per nucleon, B_ϵ/A, as a function of atomic mass number A for the most stable isobar of A.

For nuclei with mass of $A \lesssim 60$, we can fuse two light nuclei and energy is released.

In particular, the energy requirement for fusion is

$$B_\epsilon(1) + B_\epsilon(2) < B_\epsilon(3)$$

Example: The triple-alpha reaction ($3\alpha \rightarrow ^{12}\mathrm{C}$)

Consider the fusion of 3α particles to produce a ^{12}C nucleus, written to account for total binding energy,

$$\begin{aligned} 3 \times {}^4\mathrm{He} &\rightarrow {}^{12}\mathrm{C} \\ 3 \times {}^4\mathrm{He}\ (\text{in MeV}) - {}^{12}\mathrm{C}\ (\text{in MeV}) &= ? \\ 3 \times [4 \times (-7.1\ \mathrm{MeV})] - 12 \times (-7.7\ \mathrm{MeV}) &= ? \\ -85.2\ \mathrm{MeV} + 92.4\ \mathrm{MeV} &= 7.2\ \mathrm{MeV} = Q - \end{aligned}$$

In practice, mass excess ($\Delta = [M - A]c^2$) tables are used to find Q-values.

Fission on the branch with A greater than about 60 achieves the same end as the above, slowly approaching a saturation value of 8 MeV.

> nuclei around the iron peak, which are the most tightly bound of all nuclei (per constituent nucleon), are not of much use as an energy source. the end is near.

Astrophysical Thermonuclear Cross Sections and Reaction Rates

Lets start by determining the cross section for the reaction $X(\alpha, \beta)Y$ denoted as

$$\sigma_{\alpha\beta} = \frac{\text{number of reactions per unit time per target } X}{\text{incident flux of projectiles } \alpha} (\mathrm{cm}^2)$$

we can define the incident projectile flux as $n_\alpha v$ such that the reaction rate per target nucleus is $n_\alpha v \sigma_{\alpha\beta}$, leading to the **total reaction rate**

$$r_{\alpha\beta} = n_\alpha n_X \sigma_{\alpha\beta}(v)(v)\ (\mathrm{cm}^{-3}\ \mathrm{s}^{-1})$$

Here, n_X is the target number density.

We can simplify this by switching to a **center-of-mass system** expression for integrating over all particles in their respective distributions provides a averaged product of cross section and velocity:

$$r_{\alpha\beta} = n_\alpha n_X \left\langle \sigma_{\alpha\beta} \right\rangle (\mathrm{cm}^{-3}\ \mathrm{s}^{-1})$$

where here we have introduced the **reduced mass** as $m = m_\alpha m_X/(m_\alpha + m_X)$.

The cross section can be expressed in the form (usually) as a function of center of mass energy $\mathcal{E}$ as

$$\sigma_{\alpha\beta} = \pi\lambda_{dB}^2 g(2\ell + 1)\frac{\Gamma_\alpha\Gamma_\beta}{\Gamma^2} f(\mathcal{E})$$

where λ_{dB}^2 is the reduced DeBroglie wavelength, $f(\mathcal{E})$ is a shape factor, $\frac{\Gamma_\alpha\Gamma_\beta}{\Gamma^2}$ is the joint probability of forming $\alpha + X$ and then $\beta + Y$ through the compound state Z^*.

> The shape factor can take two forms: **resonant** or **non-resonant**.

Resonant - Varies rapidly with energy over some interesting energy range and is strongly peaked at a resonant energy $\mathcal{E}_r$.

Nonresonant - Shape factory is constant or is slowly varying compared to other factors in the cross section. Occurs when the energy range of interest is far from $\mathcal{E}_r$ or when the reaction is intrinsically nonresonant.

Nonresonant Reactions

Nuclear reactions of major astrophysical interest are **exothermic**: they produce energy and the Q-value is positive in the nuclear energy equation.

In the classical picture, the target and projectile could never combine because the Coulomb barrier cannot be penetrated. However, quantum mechanics can allow this to occur via tunneling with some barrier penetrability factor

$$P_\ell(\mathcal{E}) \propto e^{-2\pi\eta}$$

where is the dimensionless Sommerfield factor:

$$\eta = \frac{Z_\alpha Z_X e^2}{\bar{h}v} = 0.1574 Z_\alpha Z_X \left(\frac{\mu}{\mathcal{E}}\right)^{1/2}$$

this factor depends strongly on the entrance channel kinetic energy.

This allows us to bring this together and write the non-resonant form of the cross section

$$\sigma_{\alpha\beta} = \frac{S(\mathcal{E})}{\mathcal{E}} e^{-2\pi\eta}$$

A common procedure is to extract $\sigma_{\alpha\beta}$ experimentally, to energies non less than about 100 keV and plot the **astrophysical S factor**,

$$S(\mathcal{E}) = \sigma_{\alpha\beta}\mathcal{E}e^{2\pi\eta}$$

and then extract to lower energies expected in stellar environments. This is a major source of uncertainty in stellar nuclear reaction rates.

Assuming a constanct astrophysical S factor, we can compute a numerical form for the averaged cross section:

$$\langle\sigma_{\alpha\beta}\rangle = \frac{1.6\times 10^{-15}}{\mu^{1/2}(kT)^{3/2}} S \int_0^{\inf} \exp\left[-\left(\frac{\mathcal{E}}{kT} + \frac{b}{\mathcal{E}^{1/2}}\right)\right]$$

In the above equation the integrand is called the **Gamow peak**.

The structure of the integrand in reflects the combination of two strongly competing factors.

- The barrier penetration factor contributes the second term, which increases rapidly with increasing energy
- the Maxwell–Boltzmann exponential decreases rapidly as energy increases

> The integrand thus increases as energy increases because the Coulomb barrier becomes more penetrable but, to offset that, the number of pairs of particles available for the reaction decreases in the exponential tail of the distribution.

Example Gamow Integrand

gamow_window

The integrand plotted against center-of-mass energy (in keV) for the temperatures $T_6 = 20$ and $T_6 = 22$. Here, $T_6 = T(\mathrm{K})/T \times 10^6(\mathrm{K})$.

> We note the significant dependence of temperature for the integrand and thus the nuclear reaction rate.

Once the cross section is determined the total reaction rate can be written as

$$r_{\alpha\beta} = \rho^2 N_{\mathrm{A}}^2 \frac{X_\alpha X_X}{A_\alpha A_X}\langle\sigma_{\alpha\beta}\rangle$$

and the resulting nuclear energy generation rate

$$\epsilon_{\alpha\beta} = Q\frac{r_{\alpha\beta}}{\rho}$$

Example Astrophysical S factor measurement and extrapolation

s_factor_nr

Credit: Fowler et al., 1967 The nonresonant factor S(E) for the reaction 12C (p, γ) 13N with an extrapolation to low energies.

Resonant Reactions

To capture the resonant portion of the reaction, the form is often treated as a delta function.

This leads to a form of the resonant cross-section:

$$\langle\sigma_{\alpha\beta}\rangle = \bar{h}^2\left(\frac{2\pi}{mkT}\right)^{3/2} g(2\ell+1)\frac{\Gamma_\alpha\Gamma_\beta}{\Gamma}e^{-\mathcal{E}_r/kT}$$

Further simplification assuming the integrated cross section as a delta function leads to

$$\langle\sigma_{\alpha\beta}\rangle = \left(\frac{2\pi\bar{h}^2}{mkT}\right)^{3/2}\frac{(\omega\gamma)_r}{\bar{h}}e^{-\mathcal{E}_r/kT}$$

The term $(\omega\gamma)_r$ as often tabulated and values for specific reactions can be plugged in to further reduce the cross section equation.

Lastly, the non-resonant and resonant estimates for the cross section are added together to provide the total cross section as a function of temperature.

Other Forms of Reaction Rates

Neutron Capture and the S-Process

s-process - a slow process by which excess neutrons are captured onto "seed" nuclei in the iron range of elements. can occur for example in helium shell burning in low mass stars.

r-process - a rapid process by which a rapid succession of neutron captures lead to the formation of heavier and heavier nuclei and requiring a significant amount of neutron captures. can occur in core-collapse supernovae and neutron star mergers.

Weak Interactions

Because of the strong T and ρ dependence on different reactions, it is important to also consider the half-life of the nuclei in question.

It could be the case that beta-decay, *when atomic nucleus emits a high-energy electron or positron*, is the more likely next interaction for the created nucleus.

For example:

In the ${}^{13}\mathrm{N}(p,\gamma){}^{12}\mathrm{C}$ reaction we have been following,

$$^{13}\mathrm{N} \rightarrow {}^{13}\mathrm{C} + e^{+} + \nu_e, \tau_{1/2} = 10(\mathrm{min})$$

and the decay may be the more likley channel. At higher temperatures and densities where the resonant component can contribute, and the capture reaction is comparable to the beta decay timescale.

Another example is electron capture, consider the example of

$$e^{-} + {}^{7}\mathrm{Be} \rightarrow {}^{7}\mathrm{Li} + \nu_e$$

where the reaction is the capture of a free electron or one in an atomic orbital.

These are particularly important at **high density** environments where the reduction of free electrons and reduce electron degenracy pressure support such as in the iron core of a massive star.

Electron Screening

The final consideration for modifications to the reaction rate cross section is an overall reduction to the Coloumb potential due to intervening electrons. The net result is an increase to the penetrability factor and thus the reaction rate:

$$\left\langle \sigma_{\alpha\beta} \right\rangle (\text{with screening}) = \left\langle \sigma_{\alpha\beta} \right\rangle (\text{unscreened}) e^{U_0/kT}$$

> Screening is strongly effected by the density of the stellar environment.

The Proton–Proton Chains

The first set of nuclear burning we will discuss is that of the **The Proton–Proton Chains**. A series of reactions that lead to the production of {$^{4}\mathrm{He}$}.

Proton–Proton Chain Reactions

pp_chains

In general, the different reactions and subsequent chains become **more important as the temperature increases**. Another way to visualize these reactions is via a mass number A versus charge Z.

Proton–Proton Chain Reaction Flow

pp_chains_flow

Starting from $^{1}\mathrm{H}$, the arrows for the reaction sequences in the three pp-chains all end up at $^{4}\mathrm{He}$. The slowest reaction in the chain is the pp-reaction itself, leading to it being a bottleneck and controlling the lifetime of the star on the main-sequence.

Because these rates can contribute differently, one can often define an **effective Q-value** based on weights of the contribution for each subchain:

$$Q_{\text{eff}}(pp-\text{chains}) = 13.116\left[1 + 1.412\times10^{8}(1/X-1)e^{-4.998/T_9^{1/3}}\right]\ (\text{MeV})$$

depending only on the hydrogen mass fraction, X. This allows use to also compute an effective energy generation rate:

$$\begin{aligned}\epsilon_{\text{eff}}(pp-\text{chains}) &= r_{pp}Q_{\text{eff}}/\rho &(10)\\ &\approx \frac{2.4\times10^{4}\rho X^2}{T_9^{2/3}} &(11)\end{aligned}$$

Recall from Table 1.1 in HKT, the temperature dependence for $\epsilon_{pp} \propto^4$.

The Carbon–Nitrogen–Oxygen (CNO) Cycles

Next, we consider the CNO cycles: a series of proton captures on isotopes of CNO, positron decays and ending with a proton capture to produce ^4He.

CNO Cycle Reactions

cno

Similar to the pp-chains we can determine an effective energy generation rate, for the CNO cycle, the slowest reaction (lowest reaction rate) in the sequence is the $^{14}\text{N}(p,\gamma)^{15}\text{O}$. This reaction is often referred to as the **bottleneck** reaction rate for stars that burn H via the CNO cycles.

$$\epsilon_{\text{CNO}} \approx \frac{4.4\times10^{25}\rho XZ}{T_9^{2/3}}e^{\frac{-15.228}{T_9^{1/2}}} \tag{12}$$

The temperature exponent is significantly larger than that of the pp-chains with $\nu\approx18$.

Cross over temperature for pp-chains to CNO cycles

pp_vs_sno

Helium-Burning Reactions

Helium burning in stars will begin in earnest at sufficient tempratures (> 10^8 K) via the first step in the "triple-alpha" reaction:

$$\alpha+\alpha\rightarrow^{8}\text{Be}$$

However, recall that $^8\mathrm{Be}$ has a lifetime of only 10^{-16} seconds! So, the next stage of the reaction can only proceed at sufficient T and seed $^8\mathrm{Be}$ nuclei available. The next reaction in the sequence is

$$\alpha +^8 \mathrm{Be} \rightarrow^{12} \mathrm{C} + \gamma$$

Note an intermediate step via the creation of an excited state of $^{12}\mathrm{C}^*$ which can decay back to $^8\mathrm{Be}$. The main point being that not all forward reactions will lead to creation of $^{12}\mathrm{C}$.

We can similarly determine an energy generation rate for triple-α:

$$\epsilon_{3\alpha} \approx \frac{5.1 \times 10^8 \rho^2 Y^3}{T_9^3} e^{\frac{-4.4027}{T_9}} \tag{13}$$

At a temperature of $T_8 = 1$, the temperature exponent $\nu_{3\alpha} \approx 40$!

Next, we have the $^{12}\mathrm{C}(\alpha, \gamma)^{16}\mathrm{O}$ reaction.

> “If users find that their results in a given study are sensitive to the rate of the $^{12}\mathrm{C}(\alpha, \gamma)^{16}\mathrm{O}$ reaction, then they should repeat their calculations with 0.5 times and 2 times the values recommended here.” - Fowler (1985)

This reaction rate, which operates at tempratures of around $0.02 \lesssim T_9 \lesssim 10$.

Finally, we have the $^{16}\mathrm{O}(\alpha, \gamma)^{20}\mathrm{Ne}$ reaction.

The race between how quickly $^{12}\mathrm{C}$ is produced via triple-α and how quickly it is consumed via $^{16}\mathrm{O}(\alpha, \gamma)^{20}\mathrm{Ne}$ is of significant consequence for many different subsequent stellar evolution consequence. For example, the final C/O ratio of white dwarf star can lead to different observational properties.

Carbon, Neon, and Oxygen Burning

Carbon Burning The first of these advanced burning stages is carbon-burning via the compound heavy ion reaction chains for $^{12}\mathrm{C} +^{12} \mathrm{C}$ and their exit channels.

These compound carbon reactions are following by (p, α) and (p, γ) reactions to produce primarily $^{20}\mathrm{Ne}$ and at lesser amounts $^{23}\mathrm{Na}$ and $^{24}\mathrm{Mg}$.

The energy generation rate for these two reactions is given by

$$\epsilon(^{12}\mathrm{C} +^{12} \mathrm{C}) \approx \frac{1.43 \times 10^{42} Q \eta \rho X_{12}^2}{T_9^{3/2}} e^{\frac{-84.165}{T_9^{-1/3}}} \tag{14}$$

These reactions are susceptible to strong electron screening effects.

Neon Burning Neon burning takes place via *photodisintegration* the use of high-energy photons to break up $^{20}\mathrm{Ne}$ via the $^{20}\mathrm{Ne}(\gamma,\alpha)^{16}\mathrm{O}$.

However, temperatures are also high enough to allow the reaction sequence, $^{20}\mathrm{Ne}(\alpha,\gamma)^{24}\mathrm{Mg}^{24}\mathrm{Mg}(\alpha,\gamma)^{28}\mathrm{Si}$.

> The net result of neon burning is $^{16}\mathrm{O}$, $^{24}\mathrm{Mg}$, and $^{28}\mathrm{Si}$.

Oxygen Burning Oxygen burning proceeds in a similar fashion as carbon this time with three exit channels summarized in the Table below:

Table of Carbon- and Oxygen-Burning Reactions

advanced_burning

The energy generation rate for these reactions is given by

$$\epsilon(^{16}\mathrm{O}+{}^{16}\mathrm{O}) \approx \frac{1.3\times 10^{52} Q\eta\rho X_{16}^{2}}{T_9^{2/3}} e^{\frac{-135.93}{T_9^{-1/3}}} \times e^{\left[-0.629T_9^{2/3}-0.445T_9^{4/3}+0.0103T_9^{2}\right]}$$

We note that $^{12}\mathrm{C}+^{16}\mathrm{O}$ are possible by the seed $^{12}\mathrm{C}$ is used up quickly by $^{12}\mathrm{C}+^{12}\mathrm{C}$ and the $^{12}\mathrm{C}+^{16}\mathrm{O}$ reaction is intrinsically slow.

The final result of this stage is the production of $^{28}\mathrm{Si}$, $^{30}\mathrm{Si}$, and $^{32}\mathrm{S}$ depending on the conditions in the core.

Silicon “Burning” or "Melting"

At temperatures of about $T > 3\times 10^{9}(\mathrm{K})$, many reactions are possible. More importantly, photodisintigration plays a role in Silicon burning via the path:

$$^{28}\mathrm{Si}(\gamma,\alpha)^{24\mathrm{Mg}} \rightarrow {}^{24}\mathrm{Mg}(p,\alpha)^{27}\mathrm{Al} \rightarrow {}^{27}\mathrm{Al}(\alpha,p)^{30}\mathrm{Si}$$

In this process, the photodisintegration has essentially add two neutrons to produce $^{30}\mathrm{Si}$. Many similar pathways comprise the collection of silicon burning. An example of the reaction network is shown below:

Silicon Burning Reaction Network

silicon_burning

Credit: Adapted by Clayton (1968) from Truran et al. (1966) - A sample reaction network for silicon burning that also shows the re- actions possible between nuclei in the network.

As burning advances you can reach *quasi-statistical equilibrium* (QSE) where the photodisintigration rates roughly match the capture rates. This could allow one to leverage the Saha equation at this point.

> The result of silicon burning is production of nuclei in the iron peak. For *quiescient burning*, where much time is allowed to pass, the most abundant nuceli is $^{56}\mathrm{Fe}$. For short timescales such as in CCSNe, the electron/positron decay and electron capture rates are insufficient and the primary product is $^{56}\mathrm{Ni}$.

Neutrino Emission Mechanisms

In general, neutrino absorption or scattering requires high density or neutrino energies. We can determine this value by looking at the mean free path for a neutrino as $\lambda \sim 10^{20}\mathcal{E}_\nu^{-2}/\rho\ (\mathrm{cm})$. This can typically occur in the proto-neutron star, the collapsed iron core of a massive star where nuclear densities are reached, and the neutrinos can become "trapped".

In less extreme stellar environments, we can think of neutrinos as a power drain modifying our energy equation to be, $d\mathcal{L}/dm = -\epsilon_\nu$, neutrinos remove energy from the system.

Pair Annihilation Neutrinos

Produced by the annihilation of an electron by a positron

$$e^- + e^+ \rightarrow \nu_e + \bar{\nu}_e$$

however, this reaction requires positrons. This can be accomplished at sufficient temperatures, ($kT \sim 2m_ec^2$) such that ambient photons can undergo *pair creation* (often called pair production):

$$\gamma + \gamma \Longleftrightarrow e^- + e^+$$

Photoneutrinos and Bremsstrahlung Neutrinos

Photoneutrinos Similar to electron-photon scattering but not producing a gamma ray:

$$e^- + \gamma \rightarrow e^- + \nu_e + \bar{\nu}_e$$

Bremsstrahlung (braking radiation) Neutrinos

Yields a photon when an electron is scattered off and accelerated (positive or negative) by an ion. This is an important energy loss mechanism for hot white dwarfs.

Plasma Neutrinos

In a very dense plasma, electromagnetic waves can be quantized in such a way that they behave like relativistic Bose particles with finite mass, *plasmons*. These can decay into $e^- + e^+$ or $\nu_e + \bar{\nu}_e$ pairs.

Combined Neutrino Loss Rates neu

Credit: Adapted from the calculations of Itoh and collaborators.

In-Class Assignment: Proton-Proton Chains in the Sun

Learning Goals

1. Integrate a simple proton-proton network over the Sun's lifetime (4.5 Gyr) to find approximate **equilibrium composition** in the solar core.
2. Use `solve_ivp` to integrate the network.
3. Explore the effect of **changing the network** or **core parameters** (temperature, density, composition) and compare outcomes.

Background

The proton-proton (p–p) chains fuse hydrogen into helium through multiple pathways:

- **PP-I:** $p, d, {}^{3}\mathrm{He}, {}^{4}\mathrm{He}$
- **PP-II:** adds ${}^{7}\mathrm{Li}, {}^{7}\mathrm{Be}$
- **PP-III:** adds ${}^{8}\mathrm{Be}, {}^{8}\mathrm{B}$

Energy release and reaction rates depend strongly on temperature and composition.

Creating a network

We'll use pynucastro to create a network with all of these nuclei.

```
warning: He4 was not able to be linked in TabularLibrary
warning: d was not able to be linked in TabularLibrary
warning: B8 was not able to be linked in TabularLibrary
warning: Li7 was not able to be linked in TabularLibrary
warning: p was not able to be linked in TabularLibrary
warning: He3 was not able to be linked in TabularLibrary
warning: Be8 was not able to be linked in TabularLibrary
warning: Be7 was not able to be linked in TabularLibrary
```

Approximating composition in Sun's core

We need a reasonably accurate estimate for the composition in the core of the Sun. We'll find this by integrating an initial H/He mix for 4.5 billion years--this should find an equilibrium

```
1.420092e+17
```

```
/Users/mcoughlin/opt/anaconda3/envs/ast5011/lib/python3.12/site-packages/pynucastro/r
ates/derived_rate.py:117: UserWarning: He3 partition function is not supported by tab
les: set pf = 1.0 by default
  warnings.warn(UserWarning(f'{nuc} partition function is not supported by tables: se
t pf = 1.0 by default'))
/Users/mcoughlin/opt/anaconda3/envs/ast5011/lib/python3.12/site-packages/pynucastro/r
ates/derived_rate.py:117: UserWarning: d partition function is not supported by table
s: set pf = 1.0 by default
  warnings.warn(UserWarning(f'{nuc} partition function is not supported by tables: se
t pf = 1.0 by default'))
/Users/mcoughlin/opt/anaconda3/envs/ast5011/lib/python3.12/site-packages/pynucastro/r
ates/derived_rate.py:117: UserWarning: Be7 partition function is not supported by tab
les: set pf = 1.0 by default
  warnings.warn(UserWarning(f'{nuc} partition function is not supported by tables: se
t pf = 1.0 by default'))
/Users/mcoughlin/opt/anaconda3/envs/ast5011/lib/python3.12/site-packages/pynucastro/r
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es: set pf = 1.0 by default
  warnings.warn(UserWarning(f'{nuc} partition function is not supported by tables: se
t pf = 1.0 by default'))
/Users/mcoughlin/opt/anaconda3/envs/ast5011/lib/python3.12/site-packages/pynucastro/r
ates/derived_rate.py:117: UserWarning: Li7 partition function is not supported by tab
les: set pf = 1.0 by default
  warnings.warn(UserWarning(f'{nuc} partition function is not supported by tables: se
t pf = 1.0 by default'))
```

We'll integrate this using solve_ivp

Now let's get the composition at the final time

```
array([2.81318714e-01, 2.04411172e-18, 8.77279202e-06, 7.18672513e-01,
       2.03687202e-15, 9.99888210e-12, 4.53092291e-54, 1.95666221e-21])
```

and create a pynucastro Composition from this

```
  X(p) : 0.2813187144928235
  X(d) : 2.0441117249462563e-18
  X(He3) : 8.772792020130361e-06
  X(He4) : 0.7186725127051553
  X(Li7) : 2.036872023188449e-15
  X(Be7) : 9.998882102841983e-12
  X(Be8) : 4.530922913017576e-54
  X(B8) : 1.9566622077862025e-21
```

Importance of PP-I vs. PP-II

We can evaluate the rates with this composition

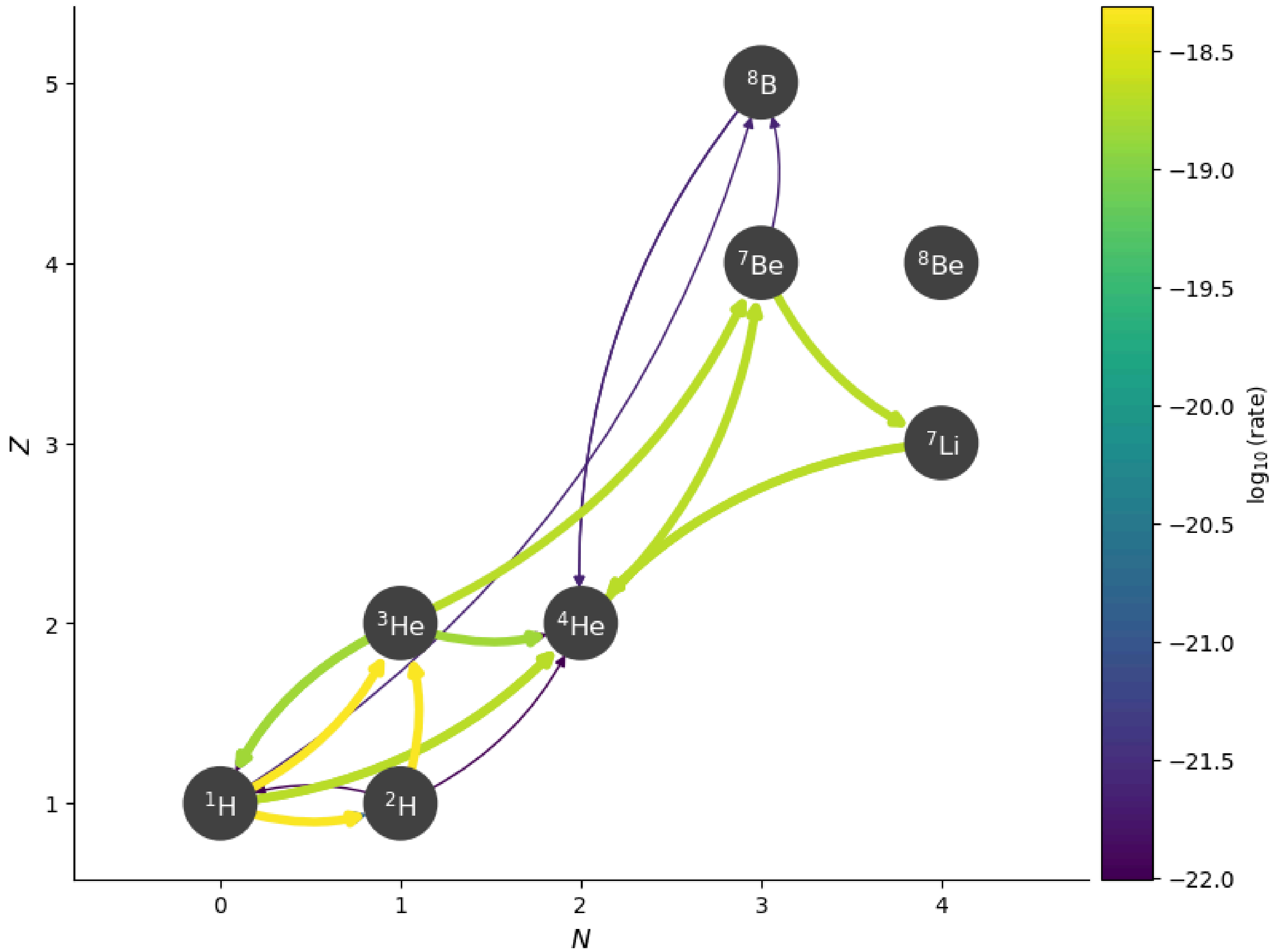
```
/Users/mcoughlin/opt/anaconda3/envs/ast5011/lib/python3.12/site-packages/pynucastro/rates/derived_rate.py:117: UserWarning: He3 partition function is not supported by tables: set pf = 1.0 by default
  warnings.warn(UserWarning(f'{nuc} partition function is not supported by tables: set pf = 1.0 by default'))
/Users/mcoughlin/opt/anaconda3/envs/ast5011/lib/python3.12/site-packages/pynucastro/rates/derived_rate.py:117: UserWarning: d partition function is not supported by tables: set pf = 1.0 by default
  warnings.warn(UserWarning(f'{nuc} partition function is not supported by tables: set pf = 1.0 by default'))
/Users/mcoughlin/opt/anaconda3/envs/ast5011/lib/python3.12/site-packages/pynucastro/rates/derived_rate.py:117: UserWarning: Be7 partition function is not supported by tables: set pf = 1.0 by default
  warnings.warn(UserWarning(f'{nuc} partition function is not supported by tables: set pf = 1.0 by default'))
/Users/mcoughlin/opt/anaconda3/envs/ast5011/lib/python3.12/site-packages/pynucastro/rates/derived_rate.py:117: UserWarning: B8 partition function is not supported by tables: set pf = 1.0 by default
  warnings.warn(UserWarning(f'{nuc} partition function is not supported by tables: set pf = 1.0 by default'))
/Users/mcoughlin/opt/anaconda3/envs/ast5011/lib/python3.12/site-packages/pynucastro/rates/derived_rate.py:117: UserWarning: Li7 partition function is not supported by tables: set pf = 1.0 by default
  warnings.warn(UserWarning(f'{nuc} partition function is not supported by tables: set pf = 1.0 by default'))
```

Let's look at just the rates that consume $^3\mathrm{He}$

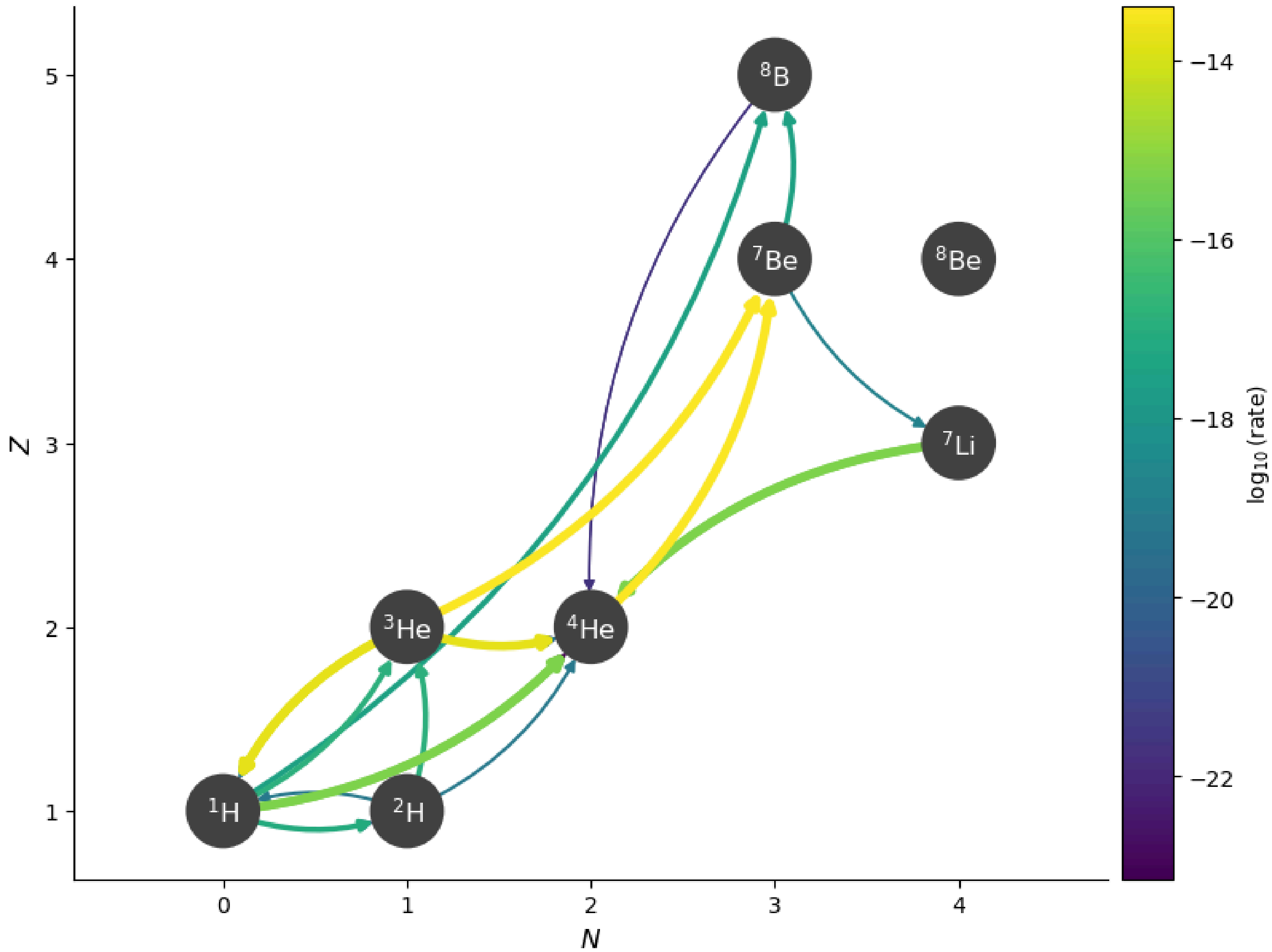
```
He3 + p ⟶ He4 + e⁺ + 𝜈        :  2.73268e-26
He3 + He4 ⟶ Be7 + 𝛾            :  1.97861e-19
He3 + H2 ⟶ p + He4             :  9.85879e-23
He3 + He3 ⟶ p + p + He4        :  1.42325e-19
Be7 + He3 ⟶ p + p + He4 + He4 :  2.64203e-39
He3 ⟶ p + H2                   :            0
```

Notice that the rate of $^3\mathrm{He}(^3\mathrm{He},pp)^4\mathrm{He}$ is only slightly slower than $^4\mathrm{He}(^3\mathrm{He},\gamma)^7\mathrm{Be}$ at this point.

```
np.float64(0.7193207261903833)
```



Now let's consider a bit hotter. We'll keep the same composition




```
np.float64(0.4169468940939774)
```

Now we see that $^4\mathrm{He}(^3\mathrm{He},\gamma)^7\mathrm{Be}$ is much faster.

Overall, as the temperature increases, we see that the $^4\mathrm{He}(^3\mathrm{He},\gamma)^7\mathrm{Be}$ rate is more active.

These rates are *strongly* dependent on the amount of $^3\mathrm{He}$ present, so we should really compute the equilibrium abundance before making the comparisons.

Experiments

Try a few variations:

1. Change the network: include or exclude PP-II/PP-III nuclei. How does the equilibrium composition and branching ratio change?
2. Change temperature: T = 1.5e7 K → T = 3.5e7 K. How do rates shift?
3. Change density: rho = 150 → rho = 300 g/cm³. How does the composition change?

> Questions:
>
> - Which reactions are most sensitive to temperature?

> - Why is He-3 abundance critical for determining branching?
> - How does this illustrate why stars burn hydrogen slowly but steadily?

▶ Answers to questions

In-class exercise: p-p chain vs. CNO

Learning Goals

1. Explore the main sequences of the p-p chain and the CNO cycle.
2. Identify rate-limiting steps in each.
3. Study temperature dependence and determine which process dominates at different stellar temperatures.

We will explore the p-p chain and the CNO cycle to learn which rates are the slowest in each and when each dominates inside of stars.

We start by getting the rates from the JINA ReacLib database. This has 80,000+ reaction rates in a standardized format.

Simple p-p chain

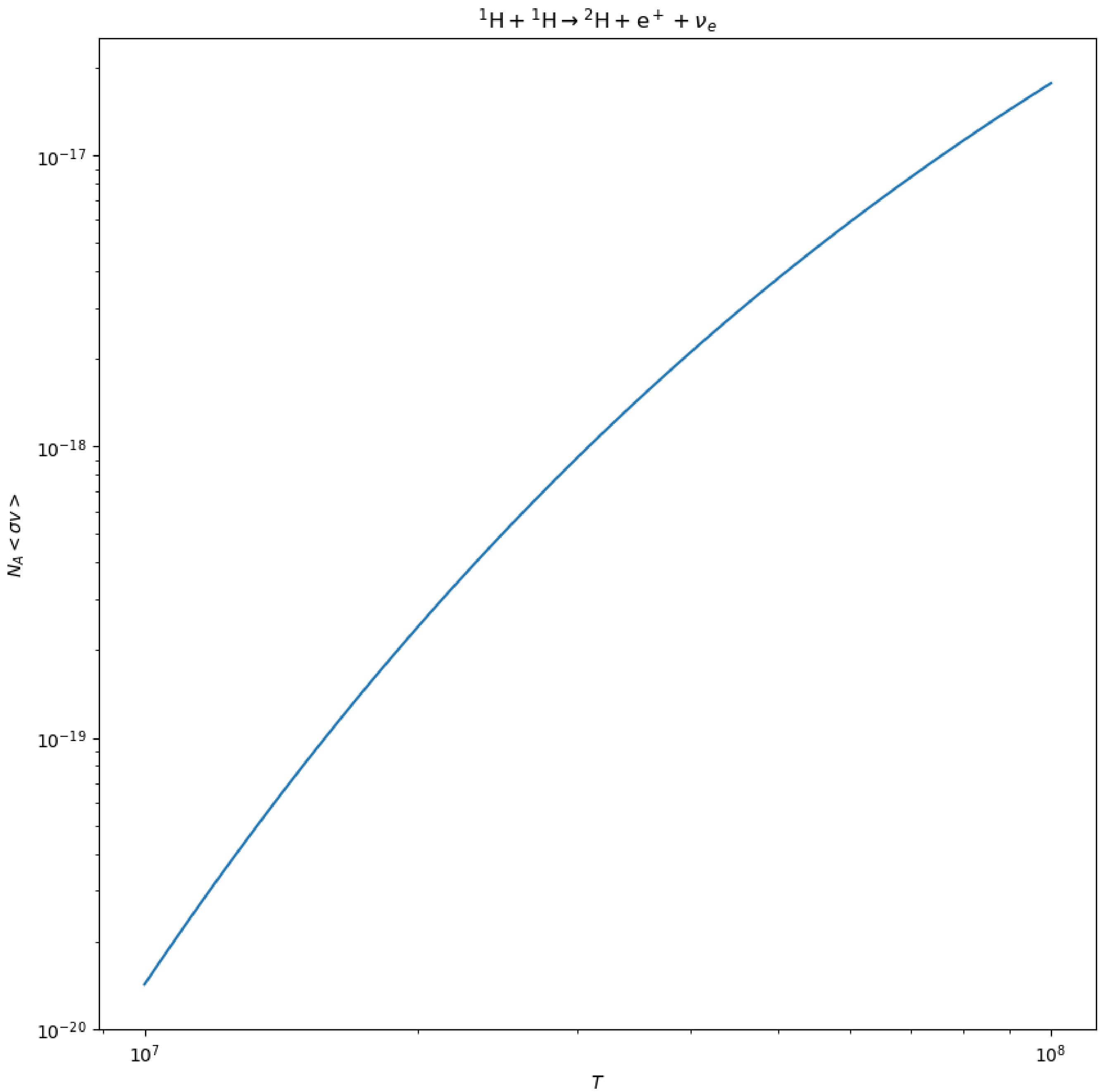
We'll start by getting the 3 main sequences that make up the p-p chain.

We can look at which rates it pulled in:

```
p + p ⟶ H2 + e⁺ + 𝜈
p + p + e⁻ ⟶ H2 + 𝜈
H2 + p ⟶ He3 + 𝛾
He3 + He3 ⟶ p + p + He4
```

Note that ReacLib actually splits the p+p rate into two. We will add them together.

We can look at the temperature dependence of one of these rates



We see that it is very temperature dependent

> **Question:** Which reactions in the p-p chain are most temperature sensitive? Which is slowest?

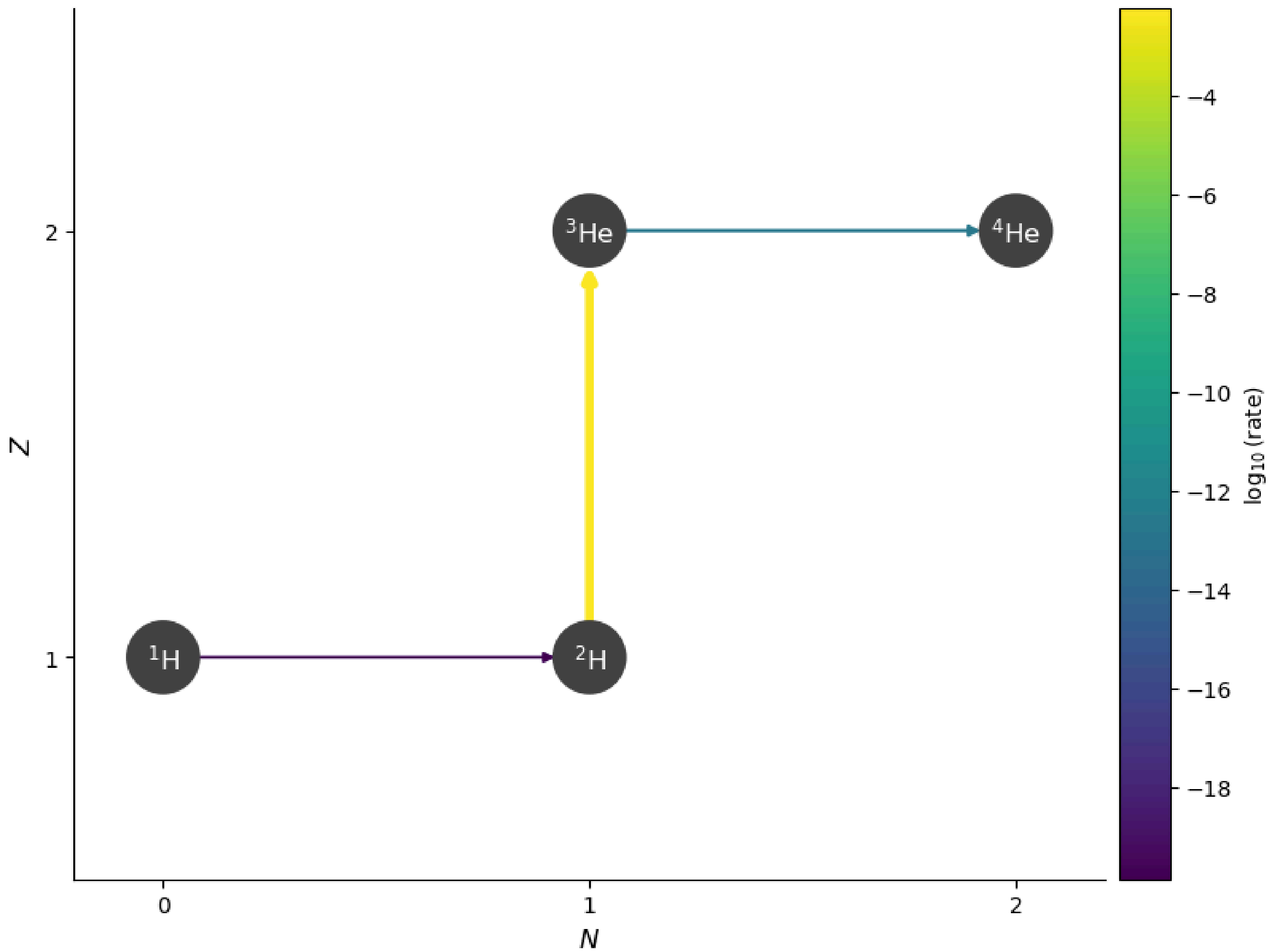
▶ Solution

Next we build our network (we'll use a `RateCollection`) with these rates

We can create a thermodynamic state approximating the center of the Sun and evaluate the rates

```
{p + p ⟶ H2 + e⁺ + 𝜈: 2.978374743269655e-18,
 p + p + e⁻ ⟶ H2 + 𝜈: 1.357778476769111e-20,
 H2 + p ⟶ He3 + 𝛾: 0.005874533290651895,
 He3 + He3 ⟶ p + p + He4: 1.849294558580789e-13}
```

We see from this that the first reaction, p + p, is by far the slowest.



> **Question:** Which reaction limits the overall p-p chain? Why is this rate so important?

▶ Solution

We can therefore approximate the p-p chain based on the slowest rate. If we express the rate as:

$$r = r_0 \left(\frac{T}{T_0} \right)^{\nu}$$

then we can find the exponent ν about some temperature T_0

```
3.9735120801889896
```

Around the conditions in the Sun, the p-p chain has a temperature dependence of $\sim T^4$

> **Question:** What does the exponent tell you about how the reaction rate changes with temperature?

▶ Solution

CNO

Now we consider the main CNO cycle. This involves 6 rates

```
C12 + p ⟶ N13 + 𝛾
C13 + p ⟶ N14 + 𝛾
N13 ⟶ C13 + e⁺ + 𝜈
N14 + p ⟶ O15 + 𝛾
N15 + p ⟶ He4 + C12
O15 ⟶ N15 + e⁺ + 𝜈
```

Again we can look evaluate each of the rates:

```
{C12 + p ⟶ N13 + 𝛾: 1.0467759242295144e-17,
 C13 + p ⟶ N14 + 𝛾: 3.176211597339995e-17,
 N13 ⟶ C13 + e⁺ + 𝜈: 2.9720852738029567e-07,
 N14 + p ⟶ O15 + 𝛾: 1.7295781984662848e-20,
 N15 + p ⟶ He4 + C12: 6.954822248932605e-16,
 O15 ⟶ N15 + e⁺ + 𝜈: 1.2625681768893481e-06}
```

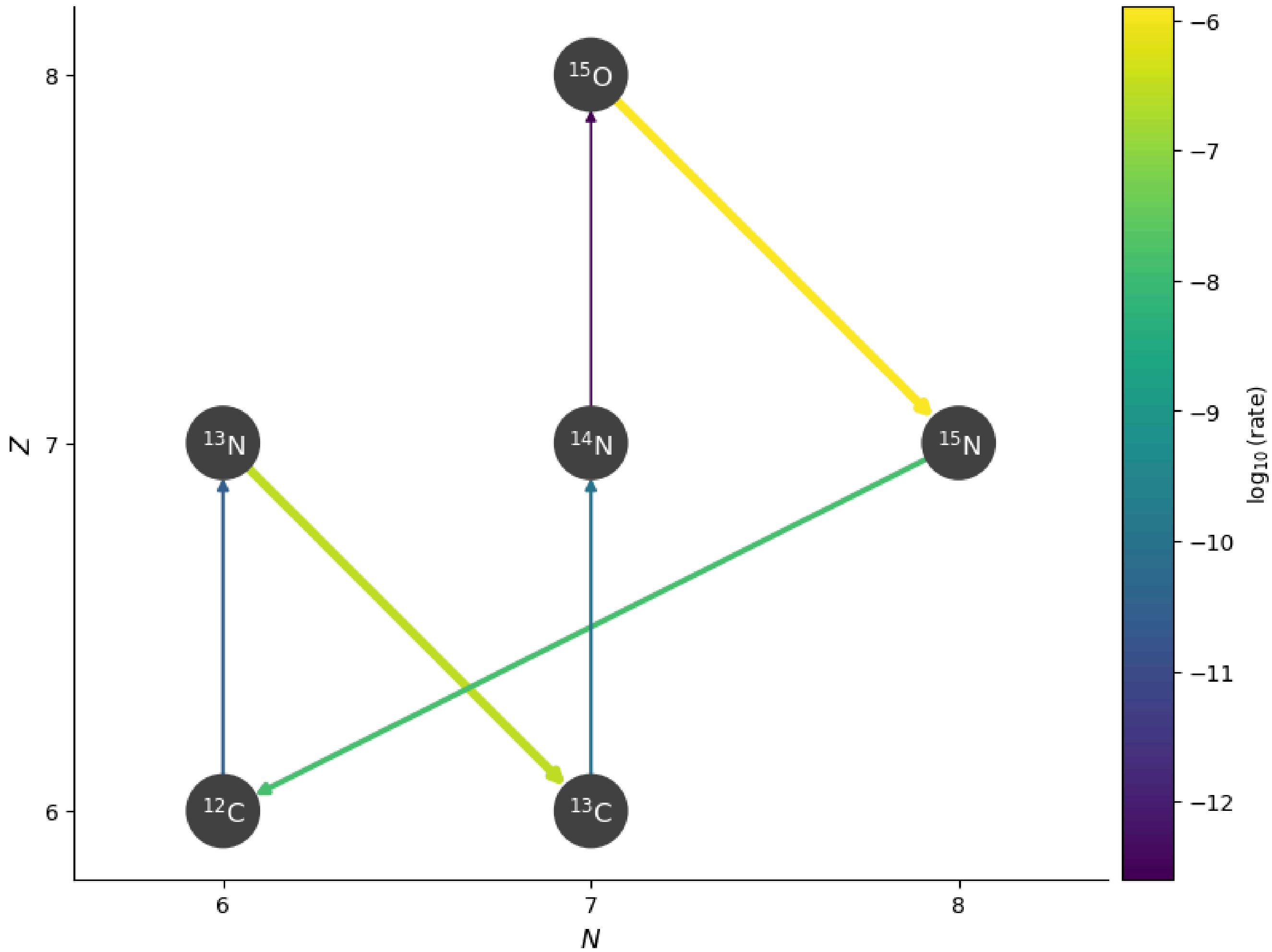
> **Question:** Which reaction is the slowest in the CNO cycle? How does this compare to the slowest p-p reaction?

▶ Solution

This is a much steeper function of temperature. Let's explore the temperature dependence around a few points

```
T = 10000000.0, nu = 22.829615097171867
T = 20000000.0, nu = 17.966372387768573
T = 30000000.0, nu = 15.601859314950396
T = 40000000.0, nu = 14.115346885210606
```

Here's what the CNO cycle looks like



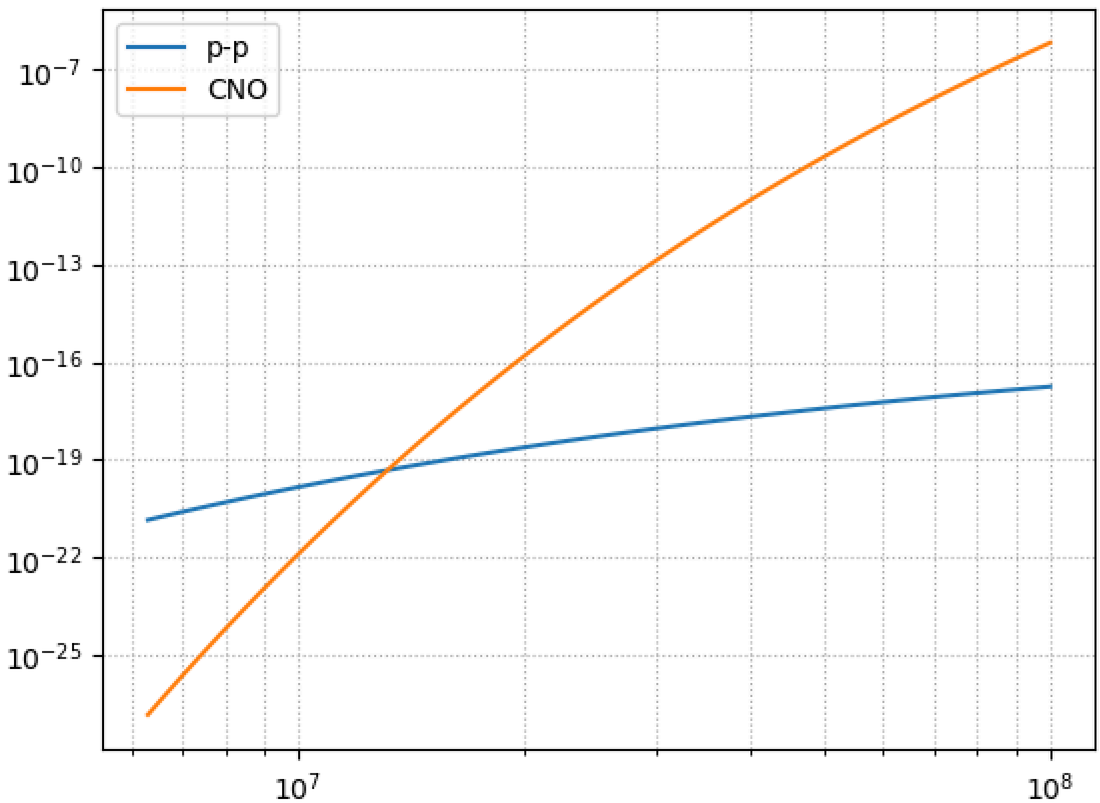
> **Question:** How does the temperature exponent for CNO compare to the p-p chain? What does this imply about which stars are powered by CNO?

▶ Solution

Comparing p-p vs. CNO

The temperature dependence suggests that if the temperature were a little higher, then CNO would go faster then p-p. We can plot the two rate-limiting rates on the same axes

```
<matplotlib.legend.Legend at 0x19bda8cb0>
```



Notice that they cross over at a temperature just hotter than the Sun's core. This means that stars more massive than the Sun (actually, about $1.5\ M_{\odot}$) are powered by CNO and less massive stars are powered by p-p. The strong T dependence of CNO will play a role in determining the structure of those more massive stars.

> **Questions:**
>
> 1. At what temperature do the p-p and CNO chains have comparable rates?
> 2. What does this tell you about which stars are powered by CNO vs. p-p?
> 3. How does the strong temperature dependence of CNO influence stellar structure?

▶ Solution

In-Class Assignment: Stellar Energy Sources

Learning Objectives

- determine dominant reaction in low mass and massive star models

Download the following model files locally.

- $1M_{\odot}$: 1m_h_dep_history.data;
- $3M_{\odot}$: 3m_h_dep_history.data;

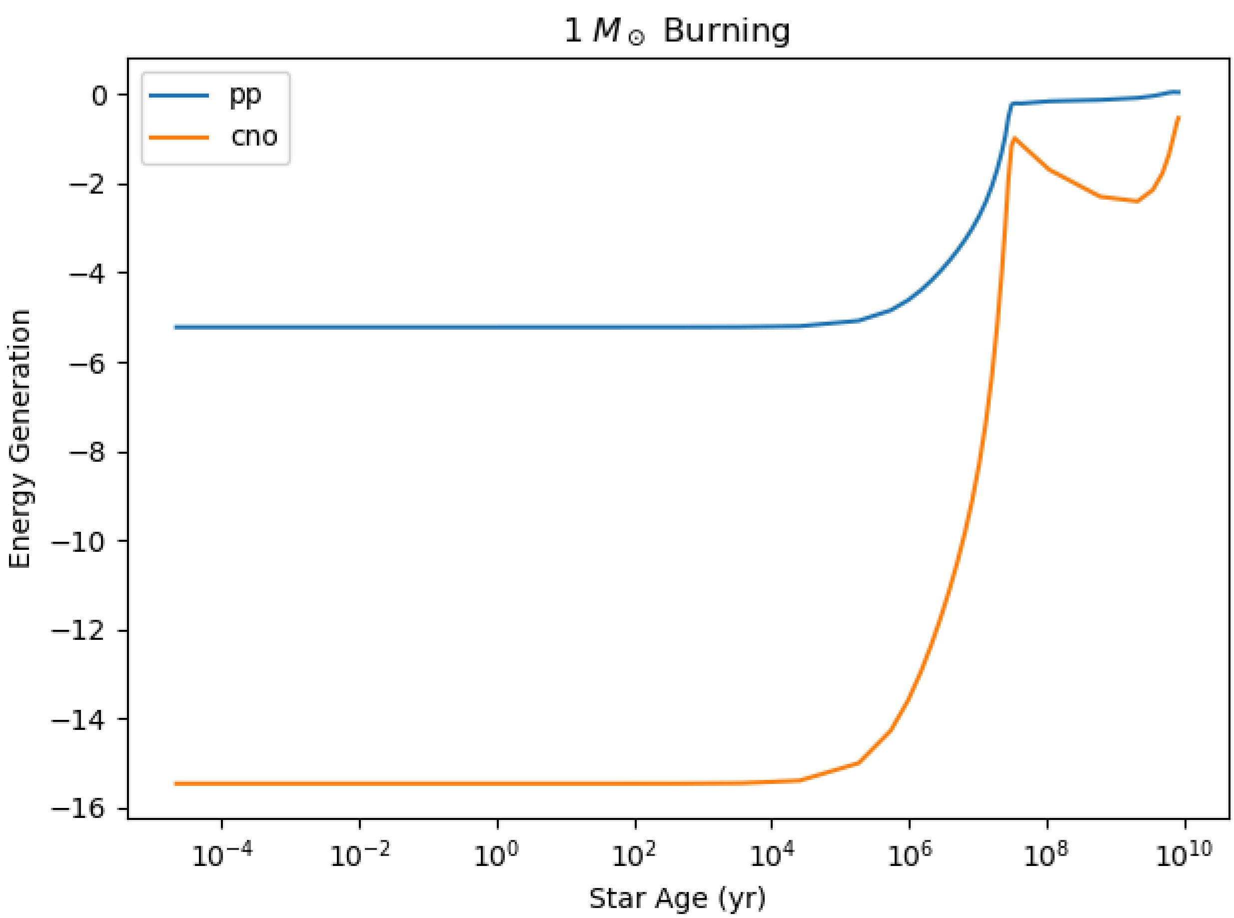
- $20M_{\odot}$: 20m_he_dep_history.data;

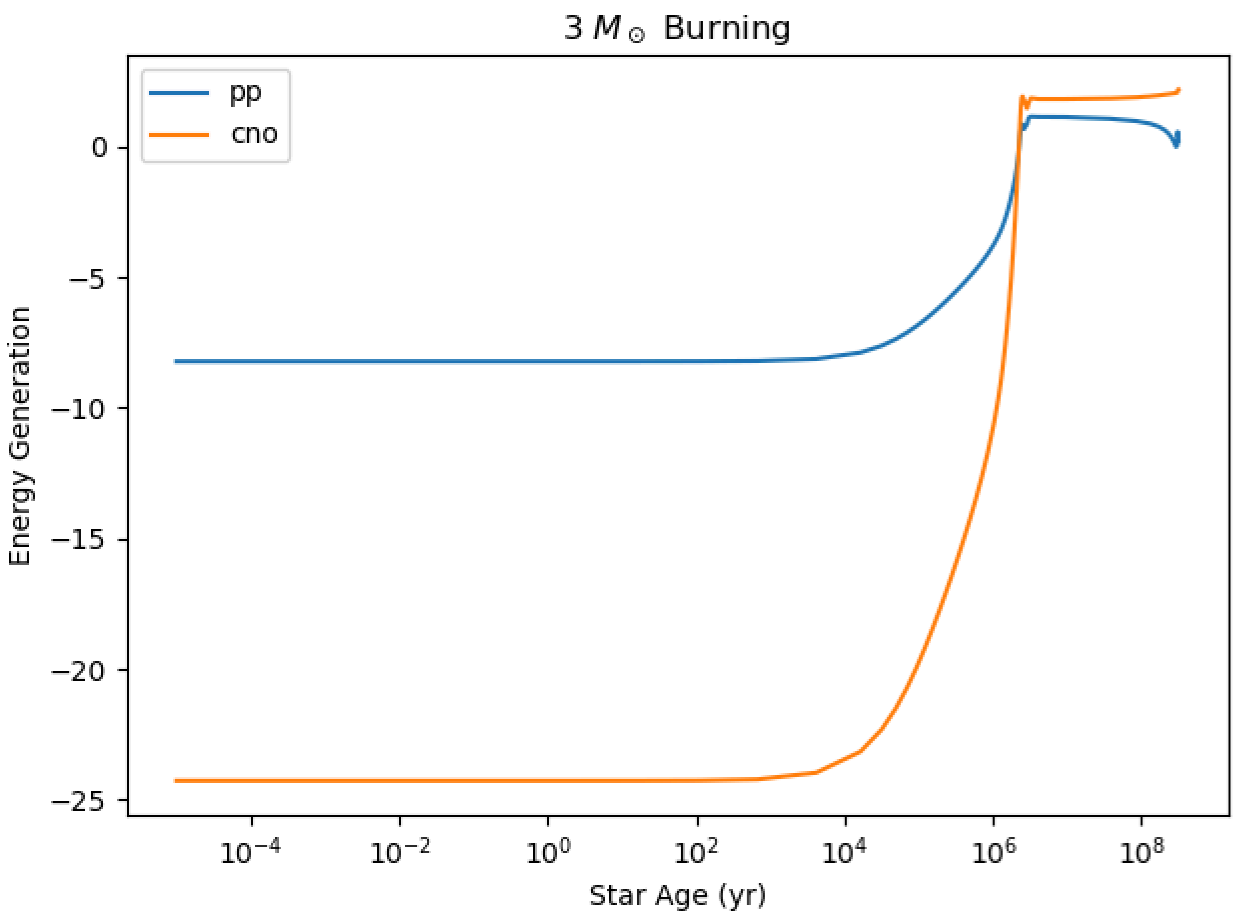
Dominant Burning Category

Using the $1M_{\odot}$ and $3M_{\odot}$ model MESA history data:

a.

1. Plot the `pp` and `cno` burning categories over time to determine which dominates in each model.

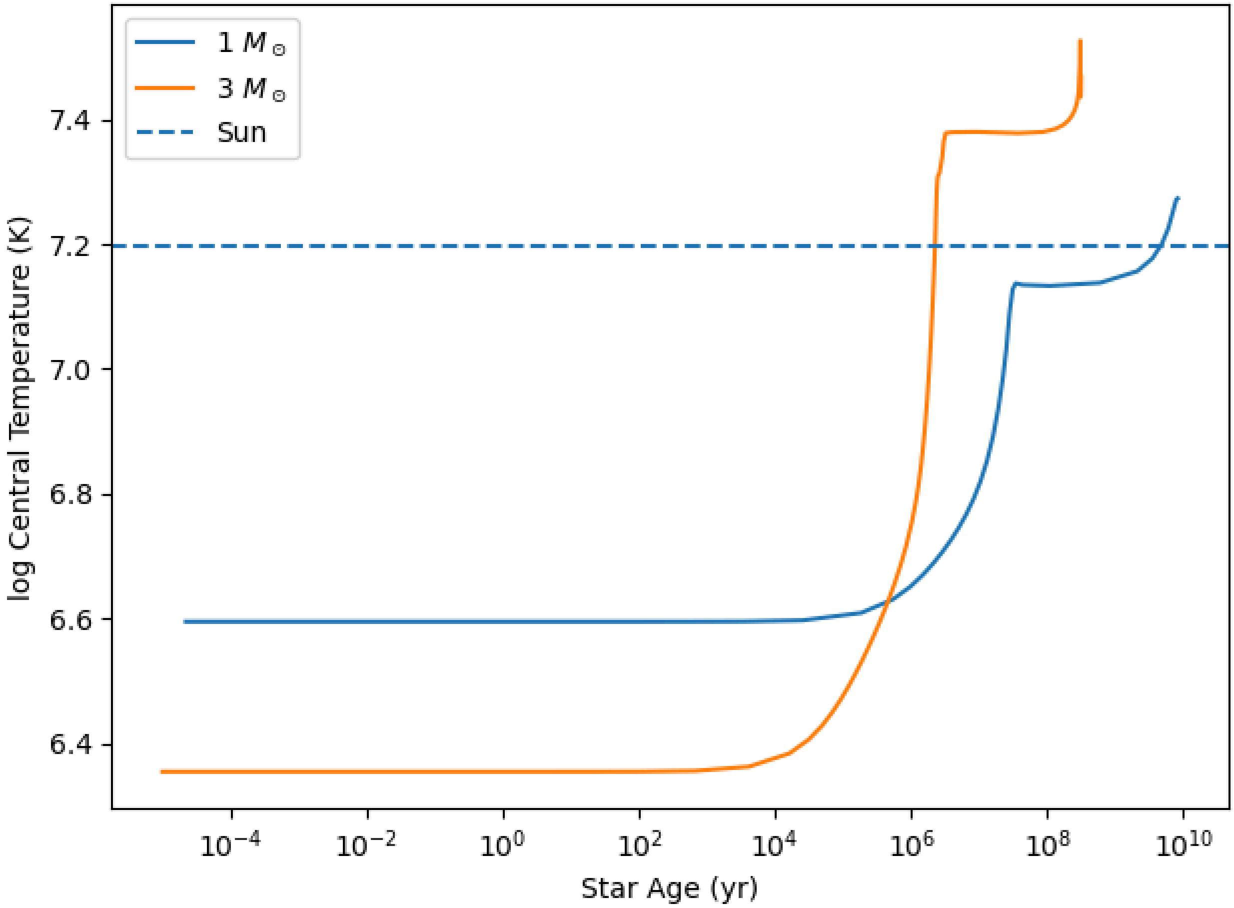




b.

1. Compare the central temperature of each models and compare with the expected central temparature of the Sun.

> Which is the dominant burning source in the Sun?

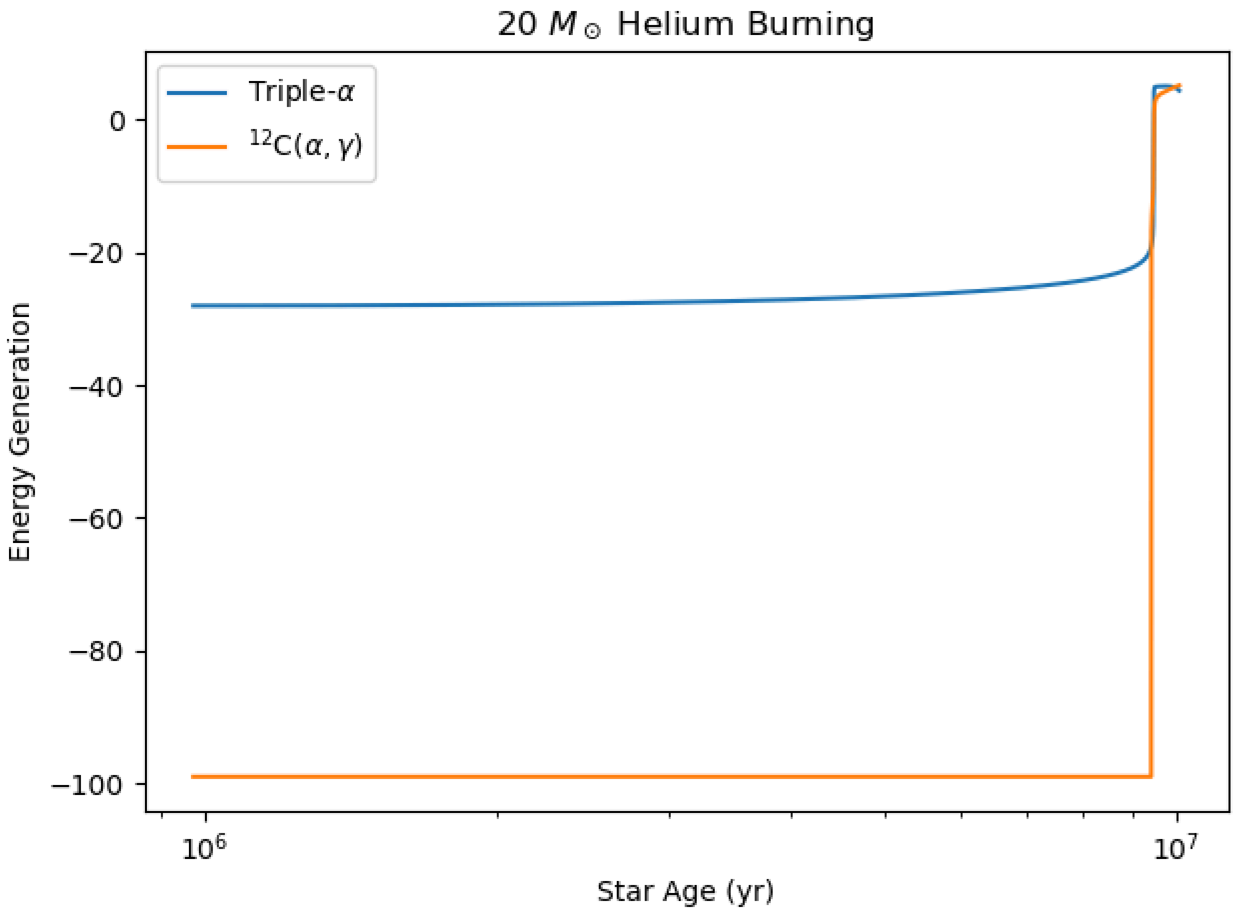


Dominant Burning in Massive Stars

Using the Massive Star model MESA history data:

C

1. Compare the `tri_alpha` and `c_alpha` categories and determine which is dominant and when.



d

1. Identify the critical central temperature, he4 and c12 mass fraction when this crossover occurs.

```
Crossover conditions:
Star age          = 9.944e+06 yr
Central T         = 2.200e+08 K
He4 mass fraction  = 1.938e-01
C12 mass fraction  = 5.629e-01
```